

On Perfect Functional Representations

Serhat Emre Coban, Yanina Y. Shkel, and Emre Telatar

Abstract—For given random variables (X, Y) , functional representation lemma states that there exists a random variable Z and a deterministic function g such that $Y = g(X, Z)$, and X is independent of Z . In other words, Z captures all the information about Y not contained in X . We refer to Z as a ‘perfect functional representation’ if it satisfies $I(Y; Z) = H(Y|X)$. We show that the existence of perfect representation depends solely on the conditional distribution $p_{Y|X}$ and define the corresponding conditional distributions as perfectly representable. We provide a necessary and sufficient condition for $p_{Y|X}$ to be perfectly representable. Finally, we cast the problem of checking perfect representability as a linear program and discuss its computational tractability.

I. INTRODUCTION

For given random variables (X, Y) , functional representation lemma [1] states that there exists a random variable Z and a deterministic function g such that $Y = g(X, Z)$, and X is independent of Z . In other words, the triple (X, Y, Z) satisfies

$$I(X; Z) = 0 \quad \text{and} \quad H(Y|X, Z) = 0. \quad (1)$$

A random variable Z that satisfies (1) is not unique. The problem of finding a Z with minimum entropy has been studied extensively in [2]–[13] and references therein. The problem of finding Z maximizing the mutual information with Y has been a key technical ingredient in information theory problems such as channel synthesis [14]–[17], and many others [14], [15]. Finally, there are many interesting connections between functional representation and information-theoretic privacy [5], [6], [18]–[26].

Of particular interest is the problem of finding a Z that is as informative as possible about Y in terms of mutual information. That is, the goal is to maximize $I(Y; Z)$ under the constraints in (1). Observe that by the chain rule,

$$I(Y; Z) = H(Y|X) - I(X; Z|Y) - H(Y|X, Z) + I(X; Z), \quad (2)$$

and by (1) we have

$$I(Y; Z) = H(Y|X) - I(X; Z|Y) \leq H(Y|X). \quad (3)$$

Hence, the maximum mutual information is trivially upper-bounded by $H(Y|X)$, with equality whenever $I(X; Z|Y) = 0$. This happens if and only if $X - Y - Z$, i.e., the triple (X, Y, Z) forms a Markov chain.

It is shown in [14], [16] that it is always possible to achieve the upper bound in (3) within a logarithmic factor. That is, there exists a Z such that

$$H(Y|X) - \log(I(X; Y) + 1) - 4 \leq I(Y; Z). \quad (4)$$

From this, we can easily see that the trivial upper bound in (3) is achieved asymptotically for product distributions. That is, applying the construction in (4) to (X^n, Y^n) gives the term $I(X^n; Z_n|Y^n)$ which is on the order of $\log n$ (and this goes to zero if we normalize by n). This problem is further studied in the context of channel synthesis in [17] where it is shown that the redundancy term $I(X^n; Z_n|Y^n)$ is $\frac{1}{2} \log n$ for (X^n, Y^n) product pairs with a so-called non-singular conditional distribution $p_{Y|X}$. Moreover, for product pairs with a singular conditional distribution, $p_{Y|X}$, the redundancy term is shown to be sub-logarithmic.

The goal of this paper is to characterize random variables for which there exists a functional representation Z such that the trivial upper bound in (3) is achieved exactly. We call such pairs (X, Y) perfectly representable and Z a perfect functional representation of the pair (X, Y) . This problem is surprisingly ubiquitous. As it is mentioned in [5], there exist a perfect functional representation in some special cases: for example, if X is a function of Y or Y is a function of X . In [14] it is shown that the term $I(X; Z|Y)$ is minimized by the Poisson functional representation whenever the support of Y is two. This problem was further studied in [18] where it is shown that if Wyner’s common information for (X, Y) is the same as the mutual information, then the pair is perfectly representable. Moreover, it could be observed from [27], [28] that perfect representability is equivalent to $H_K(G_{YX}, Y) = H(Y|X)$ where H_K denotes Körner graph entropy and the graph G_{YX} is defined as in [29]. Likewise, perfect representability is equivalent to perfect privacy in the privacy funnel, i.e., $G_{\text{PF}}(H(Y|X), X \rightarrow Y) = 0$ (see [30]), or equivalently $g_0(X; Y) = H(Y|X)$ (see [31]). Finally, this problem could be viewed as a special case of the channel synthesis problem in [17]. That is, we characterize exactly when the redundancy term in channel synthesis is zero, and not just vanishing asymptotically.

The remainder of this paper is structured as follows. In Section II, we introduce perfectly representable conditional distributions. In Section III, we provide our main results which is a necessary and sufficient condition for perfect representability. In Section IV, we present an algorithm for checking if the conditional distribution is perfectly representable.

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II. PRELIMINARIES

Let $\mathcal{X}$ and $\mathcal{Y}$ denote the support of X and Y respectively. We assume, without loss of generality, that $\mathcal{X} = \{1, 2, \dots, k\}$ and $\mathcal{Y} = \{1, 2, \dots, m\}$ for some $k, m \in \mathbb{Z}^+$.

A. Basic Definitions

Given a functional representation Z of (X, Y) such that $Y = f(X, Z)$, the vector

$$W = [f(1, Z) \ f(2, Z) \ \dots \ f(k, Z)]^T \quad (5)$$

has the following properties:

- 1) W is independent of X .
- 2) With $g(X, W) := W_X$ (i.e. X -th entry of the random vector W), we have $Y = g(X, W)$, consequently, W is also a functional representation of (X, Y) .
- 3) $W_i \sim p_{Y|X=i}$ for each $i \in \{1, 2, \dots, k\}$.
- 4) $I(Y; W) = I(Y; Z)$.

First three properties are trivial, and the last one is shown in Lemma 1 below. This observation motivates the following definition which also follows from [32, Lemma 3.22].

Definition 1. For a given $(X, Y) \sim p_{XY}$, a canonical functional representation of (X, Y) is a random vector $[W_1, W_2, \dots, W_k]^T$ independent of X such that $W_i \sim p_{Y|X=i}$ for all $i \in \{1, 2, \dots, k\}$.

Remark 1. If W is a canonical functional representation for (X, Y) , it is also a canonical functional representation for all (X', Y') with conditional distribution $p_{Y|X}$ regardless of the distribution of X' .

Lemma 1. Given an (X, Y, Z) triple that satisfies (1), there is a canonical functional representation W for the pair (X, Y) that satisfies $I(Y; W) = I(Y; Z)$.

Proof. Choose the random variable W as

$$W = [f(1, Z) \ f(2, Z) \ \dots \ f(k, Z)]^T \quad (6)$$

where f is such that $Y = f(X, Z)$. Then, W is a function of Z . By applying the equality in (3) to both Z and W , we get

$$I(Y; Z) - I(Y; W) = I(X; W|Y) - I(X; Z|Y). \quad (7)$$

Since W is a function of Z , the left hand side of (7) is non-negative and the right hand side of (7) is non-positive. Consequently, both sides are equal to zero. $\square$

Note that Lemma 1 implies that if there exists a functional representation Z of $(X, Y) \sim p_{XY}$, then there exists a canonical functional representation W that is as informative as Z about Y . Thus, it is sufficient to focus on canonical representations.

Definition 2. A conditional distribution $p_{Y|X}$ is said to be perfectly representable if there exists $p_{W|Y}$ such that W is a canonical functional representation for (X, Y) .

If $p_{XYW} = p_{W|Y}p_{Y|X}p_X$, then $X - Y - W$ is a Markov chain. If W is also a canonical functional representation, then

$I(Y; W)$ satisfies (3) with equality. From Remark 1, we see that (X, Y) being perfectly representable or not depends on $p_{Y|X}$ only. Hence, our main objective is reduced to investigating the family of conditional distributions $p_{Y|X}$ that are perfectly representable.

B. Singular Channels

Definition 3. A discrete channel $p_{Y|X}$ is called singular if, for all $y \in \mathcal{Y}$ and all $x_1, x_2 \in \mathcal{X}$ with $p_{Y|X}(y|x_1)p_{Y|X}(y|x_2) > 0$, we have $p_{Y|X}(y|x_1) = p_{Y|X}(y|x_2)$. Moreover, we call a conditional probability matrix singular if it corresponds to a singular channel.

Just like in [17], the concept of singular channels is important for our problem. As it turns out, singularity is a necessary condition for perfect representability.

Lemma 2. If a conditional distribution $p_{Y|X}$ is perfectly representable, it is singular. Moreover, if $p_{Y|X}$ is singular and $|\mathcal{X}| = 2$ or $|\mathcal{Y}| = 2$, then $p_{Y|X}$ is perfectly representable.

According to Lemma 2, singularity is a necessary condition for a channel to be perfectly representable. It is also a sufficient condition when $|\mathcal{X}| = 2$ or $|\mathcal{Y}| = 2$. However, it is not a sufficient condition in general. See Appendix A for the proof.

Example 1. For $|\mathcal{X}| = |\mathcal{Y}| = 3$, the channel

$$p_{Y|X} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix} \quad (8)$$

is singular, but not perfectly representable.

It is easy to verify that the matrix in (8) in Example 1 is singular. However, it is not perfectly representable. This can be shown in two ways. First, one may give a direct argument using Definition 2 and Definition 3; we include a formal proof as Lemma 7 in Appendix A. Second, one may appeal to Theorem 1 in Section III and easily verify that the matrix fails the necessary and sufficient condition for perfect representability stated there.

C. Total variation distance

The total variation distance of two distributions over the same alphabet is defined as

$$\text{TV}(p_{X_1}, p_{X_2}) = \frac{1}{2} \sum_{x \in \mathcal{X}} |p_{X_1}(x) - p_{X_2}(x)|. \quad (9)$$

When $|\mathcal{X}| = 2$, we can find the functional representation Z , for any p_{XY} , that achieves the maximum $I(Y; Z)$ and calculate the value of $I(Y; Z)$ in terms of the total variation distance between the conditional distributions.

Lemma 3. Given $(X, Y) \sim p_{XY}$ with $|\mathcal{X}| = 2$, define

$$r = \text{TV}(p_{Y|X=1} || p_{Y|X=2}). \quad (10)$$

The maximum value of $I(Y; Z)$ among all functional representations Z equals to $H(Y) - rH(X)$. Moreover, $rH(X)$ equals to $I(X; Y)$ if and only if $p_{Y|X}$ is singular.

This is closely connected to the fact that total variation distance gives the probability of error in a maximal coupling. See Appendix A for the proof.

III. MAIN RESULTS

In this section, we give the necessary and sufficient condition for a channel to be perfectly representable. Our approach is to define four basic operations on a singular conditional distribution matrix. Three of these operations preserve the property of being perfectly representable and three of them preserve the property of not being perfectly representable. We show that the set of all perfectly representable channels is exactly the set of channels which could be obtained via perfect-representability preserving operations from the trivial channel that maps every input to the same output.

A. Four Basic Operations

We can think of the channel $p_{Y|X}$ as a conditional probability matrix M of size $k \times m$ where $k = |\mathcal{X}|$ and $m = |\mathcal{Y}|$. Note that the i 'th column of M is $p_{Y|X}(i \cdot)$. By Lemma 2, all perfectly representable matrices must be singular. Since singularity is easy to check, we focus on singular matrices from now on.

Definition 4. For a singular conditional probability matrix M , the unique non-zero value of the non-zero entries in a column is called the "value" of that column. Moreover, a column divided by its value β is called the "pattern" of that column. So, the pattern of column i is the vector $M[:, i]/\beta$.

In Example 1, the first column of the singular matrix has value $\beta = 1/2$ and pattern $[1, 1, 0]^T$.

We define four operations on singular matrices. Note that it is straightforward to see that the result of these operations will be singular matrices as well. Let $\mathcal{S}$ denote the set of all singular matrices and $\mathcal{S}_{k,m}$ denote the set of singular matrices of size $k \times m$.

1) *Pattern preserving split (PPS)*: Let $\alpha \in (0, 1)$ and $l \in \{1, 2, \dots, m\}$. The pattern preserving split operation $A_{l,\alpha} : \mathcal{S}_{k,m} \rightarrow \mathcal{S}_{k,m+1}$ is the operation that takes a singular matrix M and splits the l 'th column into two columns (that will correspond to l 'th and $(l+1)$ 'th columns in the resulting matrix $A_{l,\alpha}(M)$) that are α and $(1-\alpha)$ times the original column.

Example 2. Let

$$M = \begin{bmatrix} 1-\beta & \beta & 0 & 0 & 0 \\ 0 & \beta & 1-\beta & 0 & 0 \\ 1-\beta & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}, \quad (11)$$

then,

$$A_{2,1/2}(M) = \begin{bmatrix} 1-\beta & \beta/2 & \beta/2 & 0 & 0 & 0 \\ 0 & \beta/2 & \beta/2 & 1-\beta & 0 & 0 \\ 1-\beta & 0 & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}. \quad (12)$$

The pattern preserving split operation is applied to the second column of M with the splitting ratio of $1/2$.

2) *Pattern preserving merge (PPM)*: The pattern preserving merge operation $B_{l_1,l_2} : \mathcal{S}_{k,m+1} \rightarrow \mathcal{S}_{k,m}$ for some distinct $l_1, l_2 \in \{1, 2, \dots, m\}$ is the operation that takes a singular matrix M and merges l_1 'th and l_2 'th columns (assuming that they have the same pattern). The new column will correspond to $\min\{l_1, l_2\}$ 'th column in the resulting matrix $B_{l_1,l_2}(M)$ that equals to the sum of these two columns (i.e. $B_{l_1,l_2}(M)[i, l] = M[i, l_1] + M[i, l_2]$ for all $i \in \{1, 2, \dots, k\}$). Then, it shifts the columns on the right side of $\max\{l_1, l_2\}$ 'th column to the left and leaves the remaining columns unchanged to get the resulting matrix.

Example 3. For the same M as in Example 2,

$$B_{4,5}(M) = \begin{bmatrix} 1-\beta & \beta & 0 & 0 \\ 0 & \beta & 1-\beta & 0 \\ 1-\beta & 0 & 0 & \beta \end{bmatrix}. \quad (13)$$

The pattern preserving merge operation is applied to the fourth and fifth columns of M .

3) *Value preserving split (VPS)*: Let $l \in [1 : m]$ and $\mathbf{b} \in \{0, 1\}^k$. The value preserving split operation $C_{l,\mathbf{b}} : \mathcal{S}_{k,m} \rightarrow \mathcal{S}_{k,m+1}$ is the operation that takes a singular matrix and splits the l 'th column (whose value is $\beta \in (0, 1)$ and whose pattern is $\mathbf{p} \in \{0, 1\}^k$) into 2 columns (that will correspond to l 'th and $(l+1)$ 'th columns in the resulting matrix $C_{l,\mathbf{b}}(M)$). Both columns of the new matrix will have value β and patterns $\mathbf{p}_1 \in \{0, 1\}^k$ and $\mathbf{p}_2 \in \{0, 1\}^k$ respectively where $\mathbf{p}_1 = \mathbf{p} \cdot \mathbf{b}$ and $\mathbf{p}_2 = \mathbf{p} \cdot (\mathbf{1}_k - \mathbf{b})$. Note that $\cdot$ represents elementwise multiplication and $\mathbf{1}_k$ represents the vector of size k all whose entries are 1.

Example 4. For the same M as in Example 2,

$$C_{1,[1,1,0]^T}(M) = \begin{bmatrix} 1-\beta & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & \beta & 1-\beta & 0 & 0 \\ 0 & 1-\beta & 0 & 0 & \beta/2 & \beta/2 \end{bmatrix}. \quad (14)$$

The value preserving split operation is applied to the first column of M with the splitting vector of $[1, 1, 0]^T$.

4) *Value preserving merge (VPM)*: Let $l_1, l_2 \in \{1, 2, \dots, m\}$ be such that the l_1 'th and l_2 'th columns of M have the same value $\beta \in (0, 1]$ and patterns $\mathbf{p}_1 \in \{0, 1\}^k$ and $\mathbf{p}_2 \in \{0, 1\}^k$ that satisfy $\mathbf{p}_1 + \mathbf{p}_2 \in \{0, 1\}^k$.

The value preserving merge (VPM) operation $D_{l_1,l_2} : \mathcal{S}_{k,m+1} \rightarrow \mathcal{S}_{k,m}$ is the operation that takes a singular matrix M and merges l_1 'th and l_2 'th columns into a single column with the pattern $\mathbf{p}_1 + \mathbf{p}_2$ and the value β (that will correspond to $\min\{l_1, l_2\}$ 'th column in the resulting matrix $D_{l_1,l_2}(M)$).

Example 5. For the same M as in Example 2,

$$D_{1,3}(M) = \begin{bmatrix} 1-\beta & \beta & 0 & 0 \\ 1-\beta & \beta & 0 & 0 \\ 1-\beta & 0 & \beta/2 & \beta/2 \end{bmatrix}. \quad (15)$$

The value preserving merge operation is applied to the first and third columns of M .

We explain how the order of these operations can be manipulated in the following lemma.

Lemma 4. *For any sequence of PPS, VPS, PPM operations, there exists an equivalent cascade that consists of a sequence of PPS operations, followed by a sequence of VPS operations, and finally a sequence of PPM operations.*

B. Perfect Representability

We are now ready to find the family of channels that are perfectly representable. We start with a channel $p_{Y|X}$ that maps every $x \in \mathcal{X}$ to a unique $a \in \mathcal{Y}$. This channel has a conditional probability matrix which is the all ones vector of length k and is clearly perfectly representable (choose $Z = a$). By Lemma 5, all matrices that we obtained from this matrix with the repeated use of pattern preserving split, pattern preserving merge, and value preserving split operations are perfectly representable. Finally, we will prove that these conditional probability matrices are the only perfectly representable conditional probability matrices.

Theorem 1. *A singular $k \times m$ conditional probability matrix M is perfectly representable if and only if it is possible to obtain the vector $\mathbf{1}_k$ from M via a repeated use of PPM, PPS, and VPM operations defined in section III-A.*

Proof. Recall that $\mathbf{1}_k$ is the all ones vector of size k and that $\mathbf{1}_k$ is perfectly representable. Thus, the "if" part of the proof is a direct result of Lemma 5 and Lemma 6 by applying the inverse of the operations mentioned in Lemma 5 to reduce M to $\mathbf{1}_k$. Example 6 illustrates this reduction.

For the "only if" part, we show that if a matrix M associated with $p_{Y|X}$ is perfectly representable, then $\mathbf{1}_k$ could be obtained from M with repeated use of PPM, VPM, and PPS operations. Since M is perfectly representable, there exists a perfect canonical functional representation W . Without loss of generality, we may assume that for each w , $p_W(w) > 0$.

First, let $Y' = (Y, W)$, so we have $Y - Y' - W$ and

$$p_{W|Y'}(w'|y, w) = \mathbb{1}\{w' = w\}, \quad (16)$$

$$p_{Y'|Y}((y, w)|y) = p_{W|Y}(w|y). \quad (17)$$

Note that W is a perfect functional representation for (X, Y') since $X - Y' - W$ and

$$Y' = (Y, W) = (g(X, W), W) =: g'(X, W). \quad (18)$$

Moreover, Y is a deterministic function of Y' and the matrix $p_{Y'|X}$ can be obtained from the matrix $p_{Y|X}$ by the PPS operation, so it is sufficient to show that $\mathbf{1}_k$ can be obtained from $p_{Y'|X}$ via PPM and VPM operations. As PPS is the inverse of PPM, and VPS is the inverse of VPM, it thus suffices to show that the matrix $p_{Y'|X}$ can be obtained from $\mathbf{1}_k$ via PPS and VPS operations. This is shown in the following paragraphs.

Next, fix $w \in \mathcal{W}$ and let

$$\mathcal{Y}'_w := \{(y, w) \in \mathcal{Y}' : y \in \mathcal{Y}\}. \quad (19)$$

Note that $\mathcal{Y}'$ can be partitioned into $\mathcal{Y}'_w$'s. Since W is a functional representation of (X, Y) , for a fixed x and fixed w , there is a unique $(y, w) \in \mathcal{Y}'_w$, namely $(y, w) = g'(x, w)$, such that

$$p_{Y'|X}((y, w)|x) > 0, \quad (20)$$

and, for any $\tilde{y} \neq y$,

$$p_{Y'|X}((\tilde{y}, w)|x) = 0. \quad (21)$$

Moreover,

$$p_{Y'|X}(g'(x, w)|x) = p_W(w) > 0. \quad (22)$$

Hence, $p_{Y'|X}(g'(x, w)|x)$ does not depend on x .

Finally, for each $y' \in \mathcal{Y}'_w$, let $S_{y'}$ be the set

$$S_{y'} = \{x \in \{1, 2, \dots, k\} : g'(x, w) = y'\}. \quad (23)$$

Since the set

$$\{p_{Y'|X}(y'|x) \cdot p_{W|Y'}(w|y') : y' \in \mathcal{Y}'\} \quad (24)$$

has exactly one non-zero element for any (x, w) pair, for each w , the collection $\{S_{y'} : y' \in \mathcal{Y}'_w\}$ is a partition of $\{1, 2, \dots, k\}$. Now for each $y' \in \mathcal{Y}'_w$, let $\mathbf{b}_{y'}$ be a vector of size k whose j -th element is $\mathbb{1}\{j \in S_{y'}\}$. We can apply PPS operation to $\mathbf{1}_k$ with coefficients $c_w = p_W(w)$, and get vectors with patterns $\mathbf{1}_k$ and values c_w for all $w \in \mathcal{W}$. Then, for all $w \in \mathcal{W}$, we can apply VPS operation to the column with pattern $\mathbf{1}_k$ and value c_w with coefficients $\mathbf{b}_{y'}$ and get $p_{Y'|X}$ as illustrated in Figure 1.

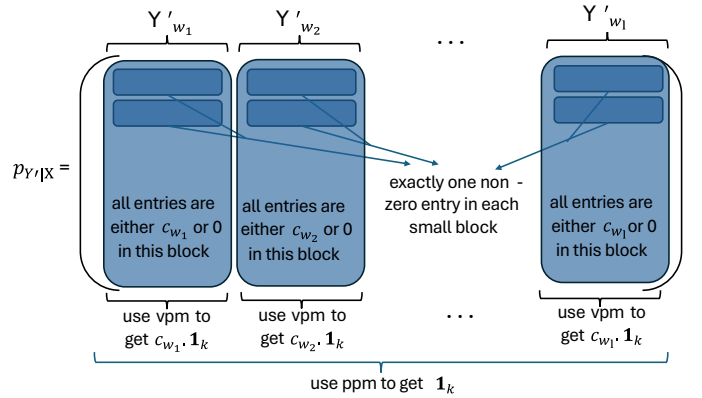


Fig. 1. Using VPM and PPM to obtain $\mathbf{1}_k$ from $p_{Y'|X}$

□

Lemma 5. *The result of applying PPM, PPS, and VPS operations described in section III-A to a perfectly representable channel will also be a perfectly representable channel.*

Since PPM and PPS operations are inverses of each other, the result of applying PPM and PPS operations to a channel that is not perfectly representable will also be a channel that is not perfectly representable. Moreover, since VPM and VPS operations are inverses of each other, the result of applying

VPM operation to a channel that is not perfectly representable will also be a channel that is not perfectly representable. Thus, Lemma 6 is a direct result of Lemma 5.

Lemma 6. *The result of applying PPM, PPS, and VPM operations described in section III-A to a channel that is not perfectly representable will also be a channel that is not perfectly representable.*

Example 6. *The following example illustrates how a representable matrix can be reduced to $\mathbf{1}_k$ using the inverse operations in Theorem 1. Consider*

$$M = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \end{bmatrix}. \quad (25)$$

Equivalently, writing the columns in terms of their values and patterns,

$$M = \left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]. \quad (26)$$

We now reduce M to $\mathbf{1}_3$.

First, apply VPM to the second and third columns:

$$\left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right] \quad (27)$$

$$\xrightarrow{\text{vpm}} \left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]. \quad (28)$$

Next, apply PPM to the two columns with pattern $\mathbf{1}_3$:

$$\left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right] \quad (29)$$

$$\xrightarrow{\text{ppm}} \left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]. \quad (30)$$

Then, apply VPM to the first two columns:

$$\left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right] \xrightarrow{\text{vpm}} \left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]. \quad (31)$$

Finally, apply PPM:

$$\left[\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right] \xrightarrow{\text{ppm}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{1}_3. \quad (32)$$

Hence, by Theorem 1, the matrix M is perfectly representable.

Remark 2. *By the equivalences noted in the introduction, Theorem 1 also characterizes exactly when*

$$H_K(G_{Y|X}, Y) = H(Y|X), \text{ (see [27]–[29])},$$

$$G_{\text{PF}}(H(Y|X), X \rightarrow Y) = 0, \text{ (see [30])},$$

$$g_0(X; Y) = H(Y|X), \text{ (see [31])}.$$

Equivalently, it characterizes exactly when the redundancy term in channel synthesis is zero (see [17]). These relations are explained in more detail in Section VI.

IV. ALGORITHMIC RESULTS

In this section, we will focus on the problem of checking if a conditional distribution $p_{Y|X}$ is perfectly representable.

Note that checking whether a matrix is a singular conditional probability matrix or not can be done in linear time. Thus, we can assume that we are working with singular conditional probability matrices. A matrix M is obtainable from $\mathbf{1}_k$ by PPS, PPM, VPS operations if and only if the matrix M' that is obtained from M by merging all columns with the same pattern is obtainable from $\mathbf{1}_k$ by PPS, PPM, VPS operations. Therefore, we can assume that each column of M has a distinct pattern. That is, it has $m \leq 2^k - 1$ columns with distinct patterns $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m$ with values $v_1, v_2, \dots, v_m$ in $[0, 1]$. Thus, our singular conditional probability matrix could be solely defined by the vector of values $\mathbf{v}$ and patterns $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m$.

A. Primal LP

Consider the all possible sets of binary vectors of length k (other than all zeros vector) that add up to $\mathbf{1}_k$. Note that the addition here is in real numbers arithmetic, not modulo 2 arithmetics. Let's call these sets as $S_1, S_2, \dots, S_R$ where R equals to the k -th Bell number which is the number of set partitions of a k -element set [33].

Example 7. *For $k = 3$, R will equal to 5 and our sets will be $S_1 = \{[1, 1, 1]^T\}$, $S_2 = \{[1, 1, 0]^T, [0, 0, 1]^T\}$, ..., $S_5 = \{[1, 0, 0]^T, [0, 1, 0]^T, [0, 0, 1]^T\}$.*

The question is whether we can assign some weights $x_1, x_2, \dots, x_R$ in $[0, 1]$ to these sets such that

$$\sum_{j: \mathbf{p}_i \in S_j} x_j = v_i, \quad \forall i \in \{1, 2, \dots, m\} \quad (33)$$

Assigning such weights $x_1, x_2, \dots, x_R$ to the sets $S_1, S_2, \dots, S_R$ is equivalent to:

1) First, from $\mathbf{1}_k$, obtain the matrix with columns whose values are $x_1, x_2, \dots, x_R$ and patterns are all $\mathbf{1}_k$ by PPS operation.

2) Then, split the i -th column with respect to S_i by VPS operation for all $i \in \{1, 2, \dots, R\}$.

3) Finally, merge all columns with the same pattern by PPM.

Then, by Lemma 4, doing any cascade of PPS, VPS, and PPM operations is equivalent to assigning weights $x_1, x_2, \dots, x_R$ to the sets $S_1, S_2, \dots, S_R$.

Now, let's define the matrix $H \in \{0, 1\}^{m \times R}$ such that $H_{ij} = \mathbb{1}\{\mathbf{p}_i \in S_j\}$.

Example 8. For $\mathbf{p}_1 = [1, 0, 0]^T, \mathbf{p}_2 = [0, 1, 0]^T, \dots, \mathbf{p}_6 = [0, 1, 1]^T, \mathbf{p}_7 = [1, 1, 1]^T$, and $k = 3$, H can be calculated as

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (34)$$

Assigning weights $\mathbf{x} = [x_1, x_2, \dots, x_R]^T$ to the sets $S_1, S_2, \dots, S_R$ such that $\sum_{j: \mathbf{p}_i \in S_j} x_j = v_i, \forall i \in \{1, 2, \dots, m\}$ is equivalent to the existence of an $\mathbf{x}$ satisfying $H\mathbf{x} = \mathbf{v}$. Therefore, our singular conditional probability matrix is perfectly representable if and only if there exists an $\mathbf{x} \in [0, 1]^R$ satisfying $H\mathbf{x} = \mathbf{v}$ where $\mathbf{v}$ is the vector of values that corresponds to our matrix. Moreover, note that the entries of $\mathbf{v}$ can not be greater than 1 and H is binary matrix, so a solution $\mathbf{x}$ for this linear system $H\mathbf{x} = \mathbf{v}$ can not have an entry greater than 1 either.

The perfect representability of our matrix is determined by the existence of $\mathbf{x} \geq 0$ satisfying $H\mathbf{x} = \mathbf{v}$. We formalize this with the following Theorem.

Theorem 2. A matrix M is perfectly representable if and only if the corresponding vector $\mathbf{v}$ and the matrix H acquired by the above procedure satisfy

$$\max_{\substack{\mathbf{x} \geq 0 \\ H\mathbf{x} \leq \mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} = 1 \quad (35)$$

Thus, we can use an algorithm that solves this LP to determine whether a given matrix M is perfectly representable or not after ensuring that M is a singular conditional probability matrix. If M is not a singular conditional probability matrix, then we say that it is not perfectly representable and we do not need to solve the linear program.

B. Dual LP

We can also solve the dual LP which has m variables instead:

Corollary 1. A matrix M is perfectly representable if and only if the corresponding vector $\mathbf{v}$ and the matrix H acquired by the above procedure satisfy

$$\min_{\substack{\mathbf{y} \geq 0 \\ H^T \mathbf{y} \geq \mathbf{1}}} \mathbf{v} \cdot \mathbf{y} = 1 \quad (36)$$

Here, our variables are $\mathbf{y} = (y_1, \dots, y_m)$, thus we have $m \leq 2^k - 1$ variables.

C. Concluding Remarks

We find a necessary and sufficient condition for a conditional distribution $p_{Y|X}$ to be perfectly representable. We provide an algorithm to test for perfect representability by casting the problem as a linear program. The dual linear program has at most $m \leq 2^k - 1$ variables. Thus, while it could be exponential in the size of $\mathcal{X}$, it is linear in the size

of $\mathcal{Y}$. The number of constraints depends on the Bell number R and could be exponential in k . We leave the question of whether this linear program could be solved efficiently to the future work.

V. COMPUTATIONAL COMPLEXITY

In section IV, we have establish an algorithm to check whether a given channel transition matrix A for the channel $p_{Y|X}$ is perfectly representable or not. This algorithm creates a linear program given in Corollary 1 so that we can determine whether the matrix A is perfectly representable or not by solving this linear program. The linear program has $\mathcal{E}_Y \leq m$ variables, namely $\mathbf{y}$, and $\mathcal{E}_Y + R$ constraints where R is the k -th Bell number [33], $m = 2^k - 1$, and $k = |\mathcal{X}|$. Then, our algorithm is polynomial time with respect to the size of the linear program. However, Bell numbers are known to grow super-exponentially [34]. Thus, our algorithm's runtime is $\text{poly}(R)$ that is super-exponential in the input alphabet size $k = |\mathcal{X}|$.

Considering, the algorithm that we have suggested in the section IV is very computationally costly, we will examine the existence of some less computationally costly algorithm in this chapter. Our conjecture is that checking whether a matrix is perfectly representable or not is an NP-hard problem.

Now, we assume that we are working with singular matrices whose values are discretized, that is, they can be represented as multiples of a small positive rational number $\frac{1}{N} > 0$ where $N \in \mathbb{Z}$. Here is the step by step procedure to check perfect representability of a singular matrix M :

Procedure: Checking perfect representability of a singular matrix M

- 1) Choose a large natural number N and divide each column into columns with the value $\frac{1}{N}$ with the pattern preserving split operation. In the end, we will acquire a singular matrix whose columns' values are only $\frac{1}{N}$.
- 2) Create a hypergraph equivalent to this acquired matrix as follows. First, create a vertex for each column. Then, assign hyper-edges between the vertices which correspond to the columns that has non-zero entry for that row. For example, the graph equivalent of the conditional probability matrix M of $p_{Y|X}$ given with:

$p_{Y X}$	$Y = a$	$Y = b$	$Y = c$	$Y = d$	$Y = e$
$X = 1$	1/3	1/3	0	0	1/3
$X = 2$	1/3	0	1/3	0	1/3
$X = 3$	0	1/3	0	1/3	1/3

is a hypergraph with 5 vertices a, b, c, d , and e and with 3 multi-edges $\{a, b, e\}, \{a, c, e\}$, and $\{b, d, e\}$. Note that this will always result in an N -uniform hypergraph because there exists N many non-zero entries on each row of the matrix.

- 3) Check whether this graph is N -colorable or not. Then, by the claim 1, the graph is N -colorable iff the matrix M is perfectly representable.

In the literature, this is known as N -uniform hypergraph coloring problem. For $N \geq 3$, this problem is known to be NP-complete (see [35]).

Claim 1. A conditional distribution $p_{Y|X}$ whose column values are all $\frac{1}{N}$ for some $N \in \mathbb{N}$ is obtainable from $\mathbf{1}_k$ with repeated use of pattern preserving merge, pattern preserving split, and value preserving split operations defined in section III-A if and only if the hypergraph equivalent of $p_{Y|X}$ created according to step 2 of the procedure in section V is N -colorable.

Note that we showed the "if" part, and also showed the "only if" part under an assumption in the appendix. This claim implies that the problem of determining whether a matrix whose column values are $1/N$ is obtainable from $\mathbf{1}_k$ with repeated use of operations 1, 2, and 3 defined in section III-A or not is equivalent to checking N -colorability of an N -uniform hypergraph. Since we can put a matrix into a form that all columns values are equal with step 1 of our procedure by using pattern preserving split only, actually, checking whether a matrix (not necessarily all column values are $1/N$) is obtainable from $\mathbf{1}_k$ with repeated use of operations 1, 2, and 3 defined in section III-A or not is equivalent to checking N -colorability of an N -uniform hypergraph. Moreover, there exists a legitimate conditional probability matrix for any N -uniform hypergraph. Therefore, we need to consider the all N -uniform hypergraphs for the coloring problem which means it is an NP-complete problem for $N \geq 3$ as shown in [35].

VI. DISCUSSION ON SOME INFORMATION MEASURES

In this section, we studied the connections between Wyner's perfect representability and some information measures such as Wyner's common information, Körner's graph entropy, and Excess Functional Information.

A. Wyner's Common Information

We will denote Wyner's common information as $C(X, Y)$ and it is defined as:

$$C(X, Y) = \inf_{p_{W|XY}: X-W-Y} I(XY; W)$$

Moreover, in [5], the followings are defined:

$$g_0(p_{XY}) = \sup_{\substack{Z: I(X; Z) = 0 \\ X-Y-Z}} I(Y; Z)$$

$$h_0(p_{XY}) = \sup_{Z: I(X; Z) = 0} I(Y; Z)$$

Since $H(Y|X) = I(Y; Z) + I(X; Z|Y) + H(Y|XZ)$ when $I(X; Z) = 0$, both $g_0(p_{XY})$ and $h_0(p_{XY})$ are less than or equal to $H(Y|X)$ (equality holds if and only if $p_{Y|X}$ is perfectly representable).

In [18], it is shown that $h_0(p_{XY}) = g_0(p_{XY})$ if $C(X, Y) = I(X; Y)$. Moreover, in [5], it is shown that both $g_0(p_{XY})$ and $h_0(p_{XY})$ are equal to $H(Y|X)$ if $h_0(p_{XY}) = g_0(p_{XY})$. Combining these two results, we can say that p_{XY} is perfectly representable if $C(X, Y) = I(X; Y)$. Although $C(X, Y) =$

$I(X; Y)$ is a sufficient condition for $p_{Y|X}$ to be perfectly representable, it is not a necessary condition. As an example of a perfectly representable channel $p_{Y|X}$ for which $C(X, Y) \neq I(X; Y)$, we can consider a binary erasure channel with erasure probability less than $1/2$ with a uniform p_X . Then, $C(X, Y) = 1$ and $I(X; Y) = 1 - \epsilon$ where ϵ is the erasure probability of the binary erasure channel. Thus, $C(X, Y) = I(X; Y)$ is not a necessary condition. However, in theorem 1, we have provided the necessary and sufficient condition. As a result, we show that $C(X, Y) = I(X; Y)$ implies that the conditional distribution $p_{Y|X}$ has to be obtainable by I_k with repeated use of pattern preserving split, pattern preserving merge, and value preserving split operations defined in section III-A.

B. Excess Functional Information [14]

Given random variables $(X, Y) \sim p_{XY}$, the excess functional information is defined as follows:

$$\Psi(X \rightarrow Y) = \min_{U: I(U; X) = 0} H(Y|U) - I(X; Y)$$

Then, we can observe that $p_{Y|X}$ is perfectly representable if and only if $\Psi(X \rightarrow Y) = 0$.

C. Körner Graph Entropy [27], [28]

Let $G = (V, E)$ be a graph and let $X \sim P$ be a random vertex in V . Let $\mathcal{I}(G)$ denote the family of independent sets of G . The Körner graph entropy of G under P is

$$H(G, P) := \min I(X; Y),$$

where the minimum is over all random variables $Y \in \mathcal{I}(G)$ such that $X \in Y$ almost surely.

Now, let G_{YX} be a graph with a set of vertices $\mathcal{Y}$ and there is an edge (y_1, y_2) if $p(x, y_1)p(x, y_2) > 0$ for some x .

An alternative equivalent definition of Körner Graph Entropy of G_{YX} is given in [29] as:

$$H_K(G_{YX}, Y) = \min_{U: U-Y-X, H(Y|XU)=0} I(Y; U).$$

Then, we can observe that $p_{Y|X}$ is perfectly representable if and only if $H_K(G_{YX}, Y) = H(Y|X)$.

D. Privacy Funnel [30]

Let $(X, Y) \sim P_{XY}$ be a pair of random variables, where Y denotes a public variable that can be disclosed, and X denotes a private variable that should be protected. A privacy mechanism is defined by a randomized mapping $P_{U|Y}$ producing a released representation U , such that the Markov chain $X - Y - U$ holds.

The privacy funnel quantifies the optimal trade-off between preserving information about the public variable Y and limiting the information leakage about the private variable X . For a given disclosure level $t \geq 0$, the privacy funnel is defined as

$$G_{PF}(t, X \rightarrow Y) = \min_{U: X-Y-U, I(Y; U) \geq t} I(X; U).$$

That is, among all released representations U that retain at least t bits of information about the public variable Y , the privacy funnel characterizes the minimum possible information leakage about the private variable X .

We observe that the channel $p_{Y|X}$ is perfectly representable if and only if $G_{\text{PF}}(H(Y|X), X \rightarrow Y) = 0$.

Moreover, in [31], the maximum t for the perfect privacy is defined as

$$g_0(X; Y) = \max\{t \geq 0 : G_{\text{PF}}(t, X \rightarrow Y) = 0\}.$$

We observe that the channel $p_{Y|X}$ is perfectly representable if and only if $g_0(X; Y) = H(Y|X)$.

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APPENDIX A

Proof of Lemma 2. We have three statements to prove:

1) Perfectly representable channels are singular.

Assume that $p_{Y|X}$ is perfectly representable. Then there exists a random vector $W = [W_1, \dots, W_k]^T$ such that $I(X; W) = 0$, $Y = W_X$, and $X - Y - W$.

Fix $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $w = (w_1, \dots, w_k) \in \mathcal{Y}^k$. Since $I(X; W) = 0$ and $Y = W_X$, we have

$$\Pr[X = x, W = w, Y = y] = p_X(x) p_W(w) \mathbb{1}\{w_x = y\}. \quad (37)$$

On the other hand, since $X - Y - W$,

$$\Pr[X = x, W = w, Y = y] = p_X(x) p_{Y|X}(y|x) p_{W|Y}(w|y). \quad (38)$$

Equating (37) and (38), and using $p_X(x) > 0$ for all $x \in \mathcal{X}$, we obtain

$$p_W(w) \mathbb{1}\{w_x = y\} = p_{Y|X}(y|x) p_{W|Y}(w|y). \quad (39)$$

Now fix $y \in \mathcal{Y}$ and suppose that $p_{Y|X}(y|x_1) > 0$ and $p_{Y|X}(y|x_2) > 0$ for some $x_1, x_2 \in \mathcal{X}$. Choose any w such that $p_{W|Y}(w|y) > 0$; such a w exists because $p_{W|Y}(\cdot|y)$ is a probability distribution. Then (39) implies, for $i = 1, 2$,

$$p_W(w) \mathbb{1}\{w_{x_i} = y\} = p_{Y|X}(y|x_i) p_{W|Y}(w|y) > 0. \quad (40)$$

Hence $\mathbb{1}\{w_{x_i} = y\} = 1$, so $w_{x_1} = w_{x_2} = y$. Substituting back into (39) gives

$$p_W(w) = p_{Y|X}(y|x_i) p_{W|Y}(w|y), \quad i = 1, 2. \quad (41)$$

Therefore

$$p_{Y|X}(y|x_1) = p_{Y|X}(y|x_2). \quad (42)$$

Since this holds for every y and every pair x_1, x_2 with $p_{Y|X}(y|x_1) > 0$ and $p_{Y|X}(y|x_2) > 0$, the channel $p_{Y|X}$ is singular.

2) Singular channels with $|\mathcal{X}| = 2$ are perfectly representable.

Without loss of generality, assume $\mathcal{X} = \{1, 2\}$ and let

$$p_i(y) := p_{Y|X}(y|i), \quad i = 1, 2. \quad (43)$$

Since the channel is singular, the alphabet $\mathcal{Y}$ can be partitioned into

$$\mathcal{A} := \{y \in \mathcal{Y} : p_1(y) = p_2(y) > 0\}, \quad (44)$$

$$\mathcal{B} := \{y \in \mathcal{Y} : p_1(y) > 0, p_2(y) = 0\}, \quad (45)$$

$$\mathcal{C} := \{y \in \mathcal{Y} : p_2(y) > 0, p_1(y) = 0\}. \quad (46)$$

Let

$$\alpha := \sum_{y \in \mathcal{A}} p_1(y) = \sum_{y \in \mathcal{A}} p_2(y). \quad (47)$$

If $\alpha = 1$, then $p_1 = p_2$, and the choice $W = (U, U)$ with $U \sim p_1$ is already a perfect functional representation. So, assume $\alpha < 1$.

Define probability distributions

$$q_{\mathcal{A}}(y) := \frac{p_1(y)}{\alpha}, \quad y \in \mathcal{A}, \quad (48)$$

$$q_{\mathcal{B}}(y) := \frac{p_1(y)}{1 - \alpha}, \quad y \in \mathcal{B}, \quad (49)$$

$$q_{\mathcal{C}}(y) := \frac{p_2(y)}{1 - \alpha}, \quad y \in \mathcal{C}. \quad (50)$$

Now construct a random vector $W = (W_1, W_2)$, independent of X , as follows:

$$W = \begin{cases} (U, U), & \text{with probability } \alpha \\ (V_1, V_2), & \text{with probability } 1 - \alpha \end{cases} \quad (51)$$

where V_1 and V_2 are independent and $U \sim q_{\mathcal{A}}$, $V_1 \sim q_{\mathcal{B}}$, and $V_2 \sim q_{\mathcal{C}}$.

Let $Y := W_X$. By construction,

$$W_1 \sim p_1, \quad W_2 \sim p_2, \quad (52)$$

hence $p_{Y|X}(y|i) = p_i(y)$ for $i = 1, 2$. Thus W is a canonical functional representation of (X, Y) .

It remains to verify that $X - Y - W$. If $y \in \mathcal{A}$, then $Y = y$ implies $W = (y, y)$ almost surely. This implies $H(W|Y) = 0$, and consequently $X - Y - W$. If $y \in \mathcal{B}$, then $p_{Y|X}(y|2) = 0$, so $X = 1$ almost surely given $Y = y$. Similarly, if $y \in \mathcal{C}$, then $X = 2$ almost surely given $Y = y$. In both cases, X is completely determined by Y . This implies $H(X|Y) = 0$, and therefore

$$X - Y - W. \quad (53)$$

Hence W is a perfect functional representation, and $p_{Y|X}$ is perfectly representable.

3) Singular channels with $|\mathcal{Y}| = 2$ are perfectly representable.

Without loss of generality, let $\mathcal{Y} = \{1, 2\}$. We have 2 cases:

a) Suppose first that there exists $x_0 \in \mathcal{X}$ such that $0 < p_{Y|X}(1|x_0) < 1$. Then also $0 < p_{Y|X}(2|x_0) < 1$.

We claim that in this case all rows of $p_{Y|X}$ are identical. Indeed, fix any $x \in \mathcal{X}$. If $p_{Y|X}(1|x) = 0$, then $p_{Y|X}(2|x) = 1$, and since both $p_{Y|X}(1|x)$ and $p_{Y|X}(2|x_0)$ are positive, singularity gives

$$p_{Y|X}(0|x) = p_{Y|X}(0|x_0), \quad (54)$$

which is impossible because the left-hand side is 1 while the right-hand side is strictly less than 1. Similarly, $p_{Y|X}(1|x) = 1$ is impossible. Hence $0 < p_{Y|X}(1|x) < 1$ for all $x \in \mathcal{X}$. By singularity, whenever $p_{Y|X}(1|x)$ and $p_{Y|X}(1|x_0)$ are positive, they must be equal, so

$$p_{Y|X}(1|x) = p_{Y|X}(1|x_0), \quad \forall x \in \mathcal{X}. \quad (55)$$

Therefore,

$$p_{Y|X}(\cdot|x) = p_{Y|X}(\cdot|x_0), \quad \forall x \in \mathcal{X}. \quad (56)$$

Hence, X and Y are independent. Then, $p_{Y|X}$ is perfectly representable as Y is a perfect functional representation for (X, Y) .

- b) Assume that there exists no $x_0 \in \mathcal{X}$ such that $0 < p_{Y|X}(1|x_0) < 1$. Then, for every $x \in \mathcal{X}$, $p_{Y|X}(1|x) \in \{0, 1\}$, so the channel is deterministic.

Now, define $f : \mathcal{X} \rightarrow \mathcal{Y}$ by

$$p_{Y|X}(f(x)|x) = 1, \quad \forall x \in \mathcal{X}, \quad (57)$$

and let

$$W = (f(1), \dots, f(k)). \quad (58)$$

Then W is deterministic, hence independent of X . Also, $Y = W_X$, and since W is deterministic we trivially have $X - Y - W$. Hence, W is a perfect functional representation.

Therefore, if $p_{Y|X}$ is singular and $|\mathcal{Y}| = 2$, then $p_{Y|X}$ is perfectly representable. $\square$

Lemma 7. *Singular channels are not necessarily perfectly representable. For instance, for $|\mathcal{X}| = |\mathcal{Y}| = 3$, the channel*

$$p_{Y|X} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix} \quad (59)$$

is singular, but not perfectly representable.

Proof. It is easy to verify that the matrix in (59) is singular. However, it is not perfectly representable. To see that the channel $p_{Y|X}$ is not perfectly representable, assume for contradiction that it is perfectly representable, and let $X \sim \text{Unif}\{1, 2, 3\}$. Then there exists a perfect canonical functional representation

$$W = (W_1, W_2, W_3) \quad (60)$$

such that

$$W_1 \sim \text{Unif}\{1, 2\}, W_2 \sim \text{Unif}\{1, 3\}, W_3 \sim \text{Unif}\{2, 3\}, \quad (61)$$

and

$$X - Y - W. \quad (62)$$

Since $\Pr[W_1 = 1] = 1/2$, there exists $w = (w_1, w_2, w_3)$ in the support of W such that $w_1 = 1$. For such a w , we have

$$\Pr[Y = 1, W = w] \geq \Pr[X = 1, W = w] > 0, \quad (63)$$

because $Y = W_X$ and $w_1 = 1$. Hence

$$\Pr[W = w | Y = 1] > 0. \quad (64)$$

Now, since X is uniform and $Y = 1$ can arise only from $X = 1$ or $X = 2$, we have

$$\Pr[X = 1 | Y = 1] = \Pr[X = 2 | Y = 1] = \frac{1}{2}. \quad (65)$$

Because $X - Y - W$, conditioning further on $W = w$ cannot change this conditional distribution, so

$$\Pr[X = 1 | Y = 1, W = w] = \frac{1}{2}. \quad (66)$$

If $w_2 \neq 1$, then the event $\{Y = 1, W = w\}$ forces $X = 1$, and therefore

$$\Pr[X = 1 | Y = 1, W = w] = 1, \quad (67)$$

which is a contradiction. Thus we must have

$$w_2 = 1. \quad (68)$$

Next, since $W_3 \in \{2, 3\}$ almost surely, either $w_3 = 2$ or $w_3 = 3$.

If $w_3 = 2$, then $\{Y = 2, W = w\}$ implies $X = 3$, whereas under the uniform input distribution,

$$\Pr[X = 1 | Y = 2] = \Pr[X = 3 | Y = 2] = \frac{1}{2}. \quad (69)$$

Again this contradicts $X - Y - W$.

If $w_3 = 3$, then $\{Y = 3, W = w\}$ implies $X = 3$, whereas

$$\Pr[X = 2 | Y = 3] = \Pr[X = 3 | Y = 3] = \frac{1}{2}, \quad (70)$$

which also contradicts $X - Y - W$.

Hence no point in the support of W can satisfy $w_1 = 1$. But this is impossible because $\Pr[W_1 = 1] = 1/2$. Therefore, the channel in (59) is not perfectly representable. $\square$

Proof of Lemma 3. By Lemma 1, it is sufficient to maximize $I(Y; W)$ over all canonical functional representations

$$W = (W_1, W_2), \quad (71)$$

where

$$W_1 \sim p_{Y|X=1}, \quad W_2 \sim p_{Y|X=2}, \quad (72)$$

and W is independent of X . Such a canonical functional representation is completely determined by a coupling of W_1 and W_2 . Moreover, since $Y = W_X$, we have

$$I(Y; W) = H(Y) - H(Y|W). \quad (73)$$

Therefore, maximizing $I(Y; W)$ is equivalent to minimizing $H(Y|W)$ over all couplings of W_1 and W_2 .

Now fix a coupling of W_1 and W_2 . Then

$$H(Y|W) \quad (74)$$

$$= H(W_X|W) \quad (75)$$

$$= \sum_{w_1, w_2} p_{W_1 W_2}(w_1, w_2) H(W_X | W = (w_1, w_2)). \quad (76)$$

For each (w_1, w_2) ,

$$H(W_X | W = (w_1, w_2)) = \begin{cases} 0, & \text{if } w_1 = w_2, \\ H(X), & \text{if } w_1 \neq w_2. \end{cases} \quad (77)$$

Indeed, if $w_1 = w_2$, then W_X is deterministic given $W = (w_1, w_2)$; whereas if $w_1 \neq w_2$, then W_X is just a relabeling of X , and hence has entropy $H(X)$. Thus

$$H(Y|W) = \Pr[W_1 \neq W_2] H(X). \quad (78)$$

Hence,

$$\min H(Y|W) = \left(\min \Pr[W_1 \neq W_2] \right) H(X), \quad (79)$$

where the minimum is taken over all couplings of W_1 and W_2 .

We now show that

$$\min \Pr[W_1 \neq W_2] = \text{TV}(p_{Y|X=1}, p_{Y|X=2}) = r. \quad (80)$$

For any coupling,

$$\Pr[W_1 = W_2] = \sum_y \Pr[W_1 = y, W_2 = y] \quad (81)$$

$$\leq \sum_y \min\{p_{Y|X=1}(y), p_{Y|X=2}(y)\}. \quad (82)$$

Therefore,

$$\Pr[W_1 \neq W_2] \geq 1 - \sum_y \min\{p_{Y|X=1}(y), p_{Y|X=2}(y)\} \quad (83)$$

$$= \text{TV}(p_{Y|X=1}, p_{Y|X=2}) \quad (84)$$

$$= r. \quad (85)$$

Moreover, equality is achieved by a maximal coupling. Hence

$$\min H(Y|W) = rH(X). \quad (86)$$

Consequently,

$$\max I(Y; Z) = \max I(Y; W) = H(Y) - rH(X). \quad (87)$$

For the second claim, recall that every functional representation satisfies

$$I(Y; Z) \leq H(Y|X) = H(Y) - I(X; Y). \quad (88)$$

Since we have shown that the maximum value of $I(Y; Z)$ equals $H(Y) - rH(X)$, it follows that

$$rH(X) \geq I(X; Y), \quad (89)$$

with equality if and only if the upper bound is attained, i.e., if and only if $p_{Y|X}$ is perfectly representable. Since $|\mathcal{X}| = 2$, Lemma 2 implies that this is equivalent to $p_{Y|X}$ being singular. Therefore,

$$rH(X) = I(X; Y) \quad (90)$$

if and only if $p_{Y|X}$ is singular. $\square$

APPENDIX B

Proof of Lemma 4. We show that every adjacent out-of-order pair of operations can be rewritten so that the order is corrected. Repeating this procedure yields the desired form.

Write a column with value c and pattern $\mathbf{p}$ as $c\mathbf{p}$. Also, if a VPS splits a column with pattern $\mathbf{p}$ into two columns, we write

$$\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2, \quad (91)$$

where the sum is in ordinary real arithmetic, so $\mathbf{p}_1, \mathbf{p}_2 \in \{0, 1\}^k$ have disjoint supports.

We will use the following local interchange rules.

1) A PPM followed by a PPS can be rewritten as PPS's followed by PPM's.

If the two operations act on disjoint columns, then they commute. Otherwise, for columns $a\mathbf{p}$ and $b\mathbf{p}$,

$$a\mathbf{p}, b\mathbf{p} \xrightarrow{\text{ppm}} (a+b)\mathbf{p} \xrightarrow{\text{pps}} \alpha(a+b)\mathbf{p}, (1-\alpha)(a+b)\mathbf{p} \quad (92)$$

can be rewritten as

$$\begin{aligned} a\mathbf{p}, b\mathbf{p} &\xrightarrow{\text{pps}} \alpha a\mathbf{p}, (1-\alpha)a\mathbf{p}, b\mathbf{p} \\ &\xrightarrow{\text{pps}} \alpha a\mathbf{p}, (1-\alpha)a\mathbf{p}, \alpha b\mathbf{p}, (1-\alpha)b\mathbf{p} \\ &\xrightarrow{\text{ppm}} \alpha(a+b)\mathbf{p}, (1-\alpha)a\mathbf{p}, (1-\alpha)b\mathbf{p} \\ &\xrightarrow{\text{ppm}} \alpha(a+b)\mathbf{p}, (1-\alpha)(a+b)\mathbf{p}. \end{aligned} \quad (93)$$

2) A VPS followed by a PPS can be rewritten as a PPS, followed by VPS's, and then a PPM.

Again, disjoint operations commute. In the non-trivial case,

$$c\mathbf{p} \xrightarrow{\text{vps}} c\mathbf{p}_1, c\mathbf{p}_2 \xrightarrow{\text{pps}} \alpha c\mathbf{p}_1, (1-\alpha)c\mathbf{p}_1, c\mathbf{p}_2 \quad (94)$$

can be rewritten as

$$\begin{aligned} c\mathbf{p} &\xrightarrow{\text{pps}} \alpha c\mathbf{p}, (1-\alpha)c\mathbf{p} \\ &\xrightarrow{\text{vps}} \alpha c\mathbf{p}_1, \alpha c\mathbf{p}_2, (1-\alpha)c\mathbf{p} \\ &\xrightarrow{\text{vps}} \alpha c\mathbf{p}_1, \alpha c\mathbf{p}_2, (1-\alpha)c\mathbf{p}_1, (1-\alpha)c\mathbf{p}_2 \\ &\xrightarrow{\text{ppm}} \alpha c\mathbf{p}_1, (1-\alpha)c\mathbf{p}_1, c\mathbf{p}_2. \end{aligned} \quad (95)$$

3) A PPM followed by a VPS can be rewritten as VPS's followed by PPM's.

If the operations are disjoint, they commute. Otherwise,

$$a\mathbf{p}, b\mathbf{p} \xrightarrow{\text{ppm}} (a+b)\mathbf{p} \xrightarrow{\text{vps}} (a+b)\mathbf{p}_1, (a+b)\mathbf{p}_2 \quad (96)$$

can be rewritten as

$$\begin{aligned} a\mathbf{p}, b\mathbf{p} &\xrightarrow{\text{vps}} a\mathbf{p}_1, a\mathbf{p}_2, b\mathbf{p} \\ &\xrightarrow{\text{vps}} a\mathbf{p}_1, a\mathbf{p}_2, b\mathbf{p}_1, b\mathbf{p}_2 \\ &\xrightarrow{\text{ppm}} (a+b)\mathbf{p}_1, a\mathbf{p}_2, b\mathbf{p}_2 \\ &\xrightarrow{\text{ppm}} (a+b)\mathbf{p}_1, (a+b)\mathbf{p}_2. \end{aligned} \quad (97)$$

Now consider any finite cascade of PPS, VPS, and PPM operations. Call an adjacent pair *out of order* if it is of one of the forms

$$(\text{ppm}, \text{pps}), \quad (\text{vps}, \text{pps}), \quad (\text{ppm}, \text{vps}). \quad (98)$$

By the three interchange rules above, each such adjacent out-of-order pair can be replaced by an equivalent finite cascade in which the relative order is corrected. Since the original cascade is finite, repeating this procedure eventually removes all out-of-order adjacent pairs.

At that point, no PPS appears to the right of a VPS or a PPM, and no VPS appears to the right of a PPM. Hence the resulting equivalent cascade consists of a sequence of PPS operations, followed by a sequence of VPS operations, and finally a sequence of PPM operations. $\square$

Proof of Lemma 5. Let M be perfectly representable, and let

$$W = (W_1, \dots, W_k) \quad (99)$$

be a perfect canonical functional representation of the corresponding pair (X, Y) . Thus,

$$I(X; W) = 0, \quad Y = W_X, \quad X - Y - W, \quad (100)$$

and

$$W_i \sim p_{Y|X}(\cdot|i), \quad i \in \{1, \dots, k\}. \quad (101)$$

We show separately that each of the three operations preserves perfect representability.

1) PPS preserves perfect representability. Let $A_{l,\alpha}$ split the symbol $l \in \mathcal{Y}$ into two new symbols l_1, l_2 , with probabilities α and $1 - \alpha$, respectively. Let $B \sim \text{Bern}(\alpha)$ be independent of (X, Y, W) , and define

$$\phi_B(y) \triangleq \begin{cases} y, & y \notin \{l\}, \\ l_1, & y = l, B = 1, \\ l_2, & y = l, B = 0. \end{cases} \quad (102)$$

Now define

$$Y' \triangleq \phi_B(Y), \quad W'_i \triangleq \phi_B(W_i), \quad i \in \{1, \dots, k\}. \quad (103)$$

Then W' is independent of X since it is a function of (W, B) , which is independent of X , and

$$W'_X = \phi_B(W_X) = \phi_B(Y) = Y'. \quad (104)$$

Moreover, for each i ,

$$W'_i \sim p_{Y'|X}(\cdot|i), \quad (105)$$

where $p_{Y'|X}$ is exactly the channel obtained from $p_{Y|X}$ by applying $A_{l,\alpha}$.

It remains to show that $X - Y' - W'$. If $y' \notin \{l_1, l_2\}$, then $Y' = y'$ is equivalent to $Y = y'$, so

$$p_{W'|X,Y'}(\cdot|x, y') = p_{W'|X,Y}(\cdot|x, y') = p_{W'|Y}(\cdot|y'), \quad (106)$$

where the last equality follows from $X - Y - W$ and the independence of B . If $y' = l_1$, then $Y' = l_1$ is equivalent to $\{Y = l, B = 1\}$, and hence

$$p_{W'|X,Y'}(\cdot|x, l_1) = p_{W'|X,Y,B}(\cdot|x, l, 1) = p_{W'|Y,B}(\cdot|l, 1), \quad (107)$$

which does not depend on x . The case $y' = l_2$ is identical. Thus $X - Y' - W'$, so $A_{l,\alpha}(M)$ is perfectly representable.

2) PPM preserves perfect representability. Let B_{l_1, l_2} merge two symbols $l_1, l_2 \in \mathcal{Y}$ that have the same pattern. Since M is perfectly representable, it is singular by Lemma 2. Hence

the columns corresponding to l_1 and l_2 have values $\alpha, \beta > 0$ and the same support set

$$S \triangleq \{x \in \mathcal{X} : p_{Y|X}(l_1|x) > 0\} \quad (108)$$

$$= \{x \in \mathcal{X} : p_{Y|X}(l_2|x) > 0\} \quad (109)$$

with

$$p_{Y|X}(l_1|x) = \alpha, \quad p_{Y|X}(l_2|x) = \beta, \quad x \in S. \quad (110)$$

Let l be the merged symbol, and define

$$\psi(y) \triangleq \begin{cases} y, & y \notin \{l_1, l_2\}, \\ l, & y \in \{l_1, l_2\}. \end{cases} \quad (111)$$

Set

$$Y' \triangleq \psi(Y), \quad W'_i \triangleq \psi(W_i), \quad i \in \{1, \dots, k\}. \quad (112)$$

Then W' is independent of X , and

$$W'_X = \psi(W_X) = \psi(Y) = Y'. \quad (113)$$

Also, for each i ,

$$W'_i \sim p_{Y'|X}(\cdot|i), \quad (114)$$

where $p_{Y'|X}$ is the merged channel.

It remains to show that $X - Y' - W'$. For $y' \neq l$, the argument is the same as in the PPS case. Now consider $y' = l$. If $x \notin S$, then $\Pr[Y' = l|X = x] = 0$, so there is nothing to prove. If $x \in S$, then

$$\Pr[Y = l_1|X = x, Y' = l] = \frac{\alpha}{\alpha + \beta}, \quad (115)$$

$$\Pr[Y = l_2|X = x, Y' = l] = \frac{\beta}{\alpha + \beta}, \quad (116)$$

which do not depend on x . Therefore, for any measurable set $\mathcal{A}$,

$$\Pr[W' \in \mathcal{A}|X = x, Y' = l] \quad (117)$$

$$= \frac{\alpha}{\alpha + \beta} \Pr[W' \in \mathcal{A}|X = x, Y = l_1] + \frac{\beta}{\alpha + \beta} \Pr[W' \in \mathcal{A}|X = x, Y = l_2]. \quad (118)$$

Since W' is a function of W and $X - Y - W$,

$$\Pr[W' \in \mathcal{A}|X = x, Y = l_j] = \Pr[W' \in \mathcal{A}|Y = l_j] \quad (119)$$

for $j \in \{1, 2\}$. Hence $\Pr[W' \in \mathcal{A}|X = x, Y' = l]$ does not depend on x , and thus $X - Y' - W'$. Therefore, $B_{l_1, l_2}(M)$ is perfectly representable.

3) VPS preserves perfect representability. Let $C_{l,b}$ split the symbol $l \in \mathcal{Y}$ into two new symbols l_1, l_2 according to a fixed vector

$$\mathbf{b} = (b_1, \dots, b_k) \in \{0, 1\}^k. \quad (120)$$

For each $i \in \{1, \dots, k\}$, define

$$\eta_i(y) \triangleq \begin{cases} y, & y \neq l, \\ l_1, & y = l, b_i = 1, \\ l_2, & y = l, b_i = 0. \end{cases} \quad (121)$$

Set

$$W'_i \triangleq \eta_i(W_i), \quad i \in \{1, \dots, k\}, \quad (122)$$

and define

$$Y' \triangleq W'_X = \eta_X(Y). \quad (123)$$

Since W' is a deterministic function of W , it is independent of X . Also, $W'_X = Y'$ by construction, and for each i ,

$$W'_i \sim p_{Y'|X}(\cdot|i), \quad (124)$$

where $p_{Y'|X}$ is exactly the channel obtained from $p_{Y|X}$ by applying $C_{l,b}$.

It remains to show that $X - Y' - W'$. Let

$$S_1 \triangleq \{i : b_i = 1\}, \quad S_2 \triangleq \{i : b_i = 0\}. \quad (125)$$

If $y' \notin \{l_1, l_2\}$, then again $Y' = y'$ is equivalent to $Y = y'$, and the claim follows from $X - Y - W$. If $y' = l_1$, then $Y' = l_1$ is equivalent to $\{Y = l, X \in S_1\}$. Since $X - Y - W$, the random variable W is independent of X given $Y = l$, and therefore any function of W , in particular W' , is also independent of X given $Y = l$. Hence, for every $x \in S_1$,

$$p_{W'|X,Y'}(\cdot|x, l_1) = p_{W'|X,Y}(\cdot|x, l) = p_{W'|Y}(\cdot|l), \quad (126)$$

which does not depend on x . The case $y' = l_2$ is identical. Thus $X - Y' - W'$, and so $C_{l,b}(M)$ is perfectly representable.

Since PPS, PPM, and VPS each preserve perfect representability, the lemma follows. $\square$

Proof of Lemma 6. This is an immediate consequence of Lemma 5 and the fact that PPS and PPM are inverses of each other, while VPS and VPM are inverses of each other.

We prove the claim by contraposition. Suppose that applying one of the operations PPM, PPS, or VPM to a channel M yields a perfectly representable channel M' . Since the inverse of that operation is, respectively, PPS, PPM, or VPS, Lemma 5 implies that applying the inverse operation to M' also yields a perfectly representable channel. But applying the inverse operation to M' recovers M . Hence M must be perfectly representable.

Therefore, if M is not perfectly representable, then the result of applying PPM, PPS, or VPM to M cannot be perfectly representable. $\square$

Proof of Theorem 2. As explained in Section IV, the matrix M is perfectly representable if and only if there exists a vector $\mathbf{x} \geq 0$ such that

$$H\mathbf{x} = \mathbf{v}, \quad (127)$$

where $H \in \{0, 1\}^{m \times R}$ is defined by

$$H_{ij} = \mathbb{1}\{\mathbf{p}_i \in S_j\}, \quad (128)$$

and $\mathbf{v} \in [0, 1]^m$ is the vector of column values of M .

Thus, it suffices to show that

$$\max_{\substack{\mathbf{x} \geq 0 \\ H\mathbf{x} \leq \mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} = 1 \quad (129)$$

if and only if there exists $\mathbf{x} \geq 0$ satisfying $H\mathbf{x} = \mathbf{v}$.

Let

$$P \triangleq [\mathbf{p}_1 \ \mathbf{p}_2 \ \cdots \ \mathbf{p}_m], \quad (130)$$

so that P is a $k \times m$ matrix whose columns are the distinct nonzero binary patterns appearing in M . By construction, each set S_j is a partition of $\mathbf{1}_k$, and therefore, for every $j \in \{1, \dots, R\}$,

$$\sum_{i=1}^m H_{ij} \mathbf{p}_i = \mathbf{1}_k. \quad (131)$$

Equivalently,

$$PH = [\mathbf{1}_k \ \mathbf{1}_k \ \cdots \ \mathbf{1}_k], \quad (132)$$

a $k \times R$ matrix whose every column is $\mathbf{1}_k$.

Now let $\mathbf{x} \geq 0$ satisfy $H\mathbf{x} \leq \mathbf{v}$. Multiplying by P , we get

$$PH\mathbf{x} \leq P\mathbf{v}. \quad (133)$$

Since the rows of the conditional probability matrix M sum to 1, we have

$$P\mathbf{v} = \mathbf{1}_k. \quad (134)$$

Also, using the form of PH above,

$$PH\mathbf{x} = (\mathbf{1}_R \cdot \mathbf{x})\mathbf{1}_k. \quad (135)$$

Substituting into (133), we obtain

$$(\mathbf{1}_R \cdot \mathbf{x})\mathbf{1}_k \leq \mathbf{1}_k, \quad (136)$$

and hence

$$\mathbf{1}_R \cdot \mathbf{x} \leq 1. \quad (137)$$

Therefore,

$$\max_{\substack{\mathbf{x} \geq 0 \\ H\mathbf{x} \leq \mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} \leq 1. \quad (138)$$

If there exists $\mathbf{x} \geq 0$ such that $H\mathbf{x} = \mathbf{v}$, then this $\mathbf{x}$ is feasible and satisfies

$$(\mathbf{1}_R \cdot \mathbf{x})\mathbf{1}_k = PH\mathbf{x} = P\mathbf{v} = \mathbf{1}_k, \quad (139)$$

so $\mathbf{1}_R \cdot \mathbf{x} = 1$. Hence,

$$\max_{\substack{\mathbf{x} \geq 0 \\ H\mathbf{x} \leq \mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} = 1. \quad (140)$$

Conversely, suppose

$$\max_{\substack{\mathbf{x} \geq 0 \\ H\mathbf{x} \leq \mathbf{v}}} \mathbf{1}_R \cdot \mathbf{x} = 1, \quad (141)$$

and let $\mathbf{x}^*$ be an optimal solution. Then

$$PH\mathbf{x}^* = (\mathbf{1}_R \cdot \mathbf{x}^*)\mathbf{1}_k = \mathbf{1}_k = P\mathbf{v}. \quad (142)$$

Thus,

$$P(\mathbf{v} - H\mathbf{x}^*) = 0. \quad (143)$$

Since $\mathbf{v} - H\mathbf{x}^* \geq 0$ and every column $\mathbf{p}_i$ of P is a nonzero binary vector, the only nonnegative vector in the kernel of P is the zero vector. Therefore,

$$\mathbf{v} - H\mathbf{x}^* = 0,$$

(144)

that is,

$$H\mathbf{x}^* = \mathbf{v}. \quad (145)$$

Hence there exists $\mathbf{x} \geq 0$ such that $H\mathbf{x} = \mathbf{v}$.

□